Asmaa Shahine

Zewail City for Science and Technology

asmaashahine63@gmail.com – +20 1123838052 – **Portfolio**

Education

Bachelor of Science in Aerospace Engineering *2022 – June 2027*
Zewail City University of Science and Technology, Egypt

Minor in Artificial Intelligence *2025 – June 2027*
Zewail City University of Science and Technology, Egypt

Graduation Project

Autonomous Aerial Robotic System for NDT Inspection *2026 – Present*
Bachelor's Graduation Project – Aerospace Engineering
Currently developing a multirotor UAV integrated with a robotic arm for autonomous non-destructive testing (NDT) inspection of industrial structures, focusing on aerial manipulation, flight stability, and inspection system integration.

Professional Experience

Egyptian Space Agency Internship *Sep 2026 – Dec 2026*
Egyptian Space Agency – On-site
Internship in the aerospace and space technology field.

Aircraft Factory Internship *Aug 2026 – Sep 2026*
Arab Organization for Industrialization (AOI)
Industrial training at the Aircraft Factory.

Engine Factory Internship *Sep 2026 – Oct 2026*
Arab Organization for Industrialization (AOI)
Industrial training at the Engine Factory.

University Lab – CNC Foam Laser Cutter for Airfoil Fabrication *Jul – Oct 2025*
Mechanical Design and Motion Control Project
Designed and built a CNC foam laser cutter for airfoil fabrication,

University Lab –UAV Design, Propulsion Selection, and Flight Testing *Jul – Oct 2025*
Designed and developed a fixed-wing UAV from conceptual design to flight testing, conducting aerodynamic and stability analyses to select suitable propulsion and control components. Integrated and validated the UAV systems through successful flight trials.

Advanced Aircraft Maintenance and Gas Turbine Engine Training *Aug 2024*
Completed advanced training in aircraft maintenance and gas turbine engine systems.

Gas Turbine Engine Training *Aug 2024*
Completed technical training focused on gas turbine engines.

Aircraft Maintenance Engineering Training *Aug 2024*
Completed hands-on aircraft maintenance training covering maintenance procedures, troubleshooting, and repair practices.

Project Experience

ROTORTHON 2024 – VTOL Urban Air Mobility Design *2024*
Team Member, Aerospace Engineering Club Achieved 4th place worldwide by designing a sustainable and efficient VTOL concept focused on urban air mobility, noise reduction, safety, and environmental impact.

CFD Simulation for Gas Turbine Blade Cooling *2025*
Conducted CFD-based thermal analysis of gas turbine blade cooling using ANSYS to evaluate temperature gradients and improve cooling efficiency.

HAWT Blade CFD and Structural Analysis *2025*
Performed integrated CFD, structural stress, and modal analyses of a HAWT blade to evaluate aerodynamic performance, structural integrity, and vibration behavior, optimizing efficiency and safety.

Design and Prototyping of a 3-DOF Robotic Manipulator *2025*
Designed, simulated, and fabricated a 3-DOF robotic manipulator, integrating kinematic modeling, SolidWorks CAD, 3D printing/CNC manufacturing, and microcontroller-based motor control.

Active Suspension System Modeling and Control Design *2025*
Modeled and analyzed a quarter-car active suspension system and designed state-feedback and LQR controllers using MATLAB/Simulink to enhance stability and ride comfort. **Active Suspension**

System Modeling and Control Design *2025*
Modeled and analyzed a quarter-car active suspension system and designed state-feedback and LQR controllers using MATLAB/Simulink to enhance stability and ride comfort.

Advanced Shaft Design Project *2025*
Designed and optimized a shaft system, including material selection, design calculations, and selection of suitable gears and bearings.

Siphon System & Venturi Aerator Design *2024*
Designed and modeled a siphon system and Venturi aerator using CAD software, validating performance through simulations and physical prototypes.

Flight Dynamics, Stability Analysis, and Autopilot Control Design *2026*
Modeled the Lockheed JetStar/C-140 aircraft using NASA CR-2144 data and performed flight dynamics, stability, modal, and time-response analyses in MATLAB. Designed a pitch-angle feedback autopilot that reduced long-period settling time from 409.83 s to 88.50 s, achieving a 78.4

Thin-Walled Wing Structural Design for Utility General Aviation Aircraft *2026*
Designed a thin-walled wing structure for a utility-category aircraft, optimizing airfoil selection and wing geometry using XFLR5 and analytical calculations. Developed a CAD model with spars, stringers, ribs, and skin, and applied Prandtl's Lifting-Line Theory in MATLAB to analyze lift distribution and aerodynamic performance.

Leadership & Extra-curricular Activities

AIAA Chapter Initiative *2024*
Vice Chairperson & Founding Member, Zewail City

Hult Prize Competition *2025*
Startup Team Member

Zewail City Festival (EGYPES) *2022*
Volunteer Participant

Mathematics Online Tutor *2024 – 2026*
Independent Tutor

Aerospace Instructor – Space Point Company *2025*
Technical Instructor